

1.6 Plan: Apply the definitions of chemical and physical changes to the examples.

Solution:

- a) Not a chemical change, but a **physical change** — simply cooling returns the soup to its original form.
- b) There is a **chemical change** — cooling the toast will not “un-toast” the bread.
- c) Even though the wood is now in smaller pieces, it is still wood. There has been no change in composition, thus this is a **physical change**, and not a chemical change.
- d) This is a **chemical change** converting the wood (and air) into different substances with different compositions. The wood cannot be “unburned.”

1.19 Plan: Review the table of conversions in the Chapter or inside the back cover of the book.

Solution:

a) To convert from in^2 to cm^2 , use $\frac{(2.54 \text{ cm})^2}{(1 \text{ in})^2}$; to convert from cm^2 to m^2 , use $\frac{(1 \text{ m})^2}{(100 \text{ cm})^2}$

b) To convert from km^2 to m^2 , use $\frac{(1000 \text{ m})^2}{(1 \text{ km})^2}$; to convert from m^2 to cm^2 , use $\frac{(100 \text{ cm})^2}{(1 \text{ m})^2}$

c) This problem requires two conversion factors: one for distance and one for time. It does not matter which conversion is done first. Alternate methods may be used.

To convert distance, mi to m, use:

$$\left(\frac{1.609 \text{ km}}{1 \text{ mi}} \right) \left(\frac{1000 \text{ m}}{1 \text{ km}} \right) = 1.609 \times 10^3 \text{ m/mi}$$

To convert time, h to s, use:

$$\left(\frac{1 \text{ h}}{60 \text{ min}} \right) \left(\frac{1 \text{ min}}{60 \text{ s}} \right) = 1 \text{ h} / 3600 \text{ s}$$

Therefore, the complete conversion factor is $\left(\frac{1.609 \times 10^3 \text{ m}}{1 \text{ mi}} \right) \left(\frac{1 \text{ h}}{3600 \text{ s}} \right) = \frac{0.4469 \text{ m h}}{\text{mi s}}$. Do the units cancel

when you start with a measurement of mi/h?

d) To convert from pounds (lb) to grams (g), use $\frac{1000 \text{ g}}{2.205 \text{ lb}}$;

to convert volume from ft^3 to cm^3 use: $\left(\frac{(1 \text{ ft})^3}{(12 \text{ in})^3} \right) \left(\frac{(1 \text{ in})^3}{(2.54 \text{ cm})^3} \right) = 3.531 \times 10^{-5} \text{ ft}^3/\text{cm}^3$

1.23 Plan: Anything that increases the mass or decreases the volume will increase the density (density = $\frac{\text{mass}}{\text{volume}}$).

Solution:

- a) Density **increases**. The mass of the chlorine gas is not changed, but its volume is smaller.
- b) Density **remains the same**. Neither the mass nor the volume of the solid has changed.
- c) Density **decreases**. Water is one of the few substances that expands on freezing. The mass is constant, but the volume increases.
- d) Density **increases**. Iron, like most materials, contracts on cooling, thus the volume decreases while the mass does not change.
- e) Density **remains the same**. The water does not alter either the mass or the volume of the diamond.

1.26 Plan: Use conversion factors from the inside back cover: $10^{-12} \text{ m} = 1 \text{ pm}$; $10^{-9} \text{ m} = 1 \text{ nm}$

Solution:

$$\text{Radius} = 1430 \text{ pm} \left(\frac{10^{-12} \text{ m}}{1 \text{ pm}} \right) \left(\frac{1 \text{ nm}}{10^{-9} \text{ m}} \right) = 1.43 \text{ nm}$$

1.28 Plan: Use conversion factors: $0.01 \text{ m} = 1 \text{ cm}$, $1 \text{ inch} = 2.54 \text{ cm}$.

Solution:

$$\text{Length} = 100. \text{ m} \left(\frac{1 \text{ cm}}{0.01 \text{ m}} \right) \left(\frac{1 \text{ inch}}{2.54 \text{ cm}} \right) = 3.9370 \times 10^3 = \mathbf{3.94 \times 10^3 \text{ in}}$$

1.30 a) Plan: Use conversion factors: $(0.01 \text{ m})^2 = (1 \text{ cm})^2$; $(1 \text{ km})^2 = (1000 \text{ m})^2$

Solution:

$$20.7 \text{ cm}^2 \left(\frac{(0.01 \text{ m})^2}{(1 \text{ cm})^2} \right) \left(\frac{(1 \text{ km})^2}{(1000 \text{ m})^2} \right) = \mathbf{2.07 \times 10^{-9} \text{ km}^2}$$

b) Plan: Use conversion factor: $(1 \text{ inch})^2 = (2.54 \text{ cm})^2$

Solution:

$$20.7 \text{ cm}^2 \left(\frac{(1 \text{ in})^2}{(2.54 \text{ cm})^2} \right) \left(\frac{\$3.25}{1 \text{ in}^2} \right) = 10.4276 = \mathbf{\$10.43}$$

1.32 Plan: Use conversion factor: $1 \text{ kg} = 2.205 \text{ lb}$. The following assumes a body weight of 155 lbs. Use your own body weight in place of the 155 lbs.

Solution:

$$155 \text{ lb} \left(\frac{1 \text{ kg}}{2.205 \text{ lb}} \right) = \mathbf{70.3 \text{ kg}}$$

Answers will vary, depending on the person's mass.

1.42 Plan: Use the equations given in the text for converting between the three temperature scales.

Solution:

$$\text{a) } T (\text{in } ^\circ\text{C}) = [T (\text{in } ^\circ\text{F}) - 32] \frac{5}{9} = [68^\circ\text{F} - 32] \frac{5}{9} = \mathbf{20. ^\circ\text{C}}$$

$$T (\text{in K}) = T (\text{in } ^\circ\text{C}) + 273.15 = 20. ^\circ\text{C} + 273.15 = 293.15 = \mathbf{293 \text{ K}}$$

$$\text{b) } T (\text{in K}) = T (\text{in } ^\circ\text{C}) + 273.15 = -164^\circ\text{C} + 273.15 = 109.15 = \mathbf{109 \text{ K}}$$

$$T (\text{in } ^\circ\text{F}) = \frac{9}{5} T (\text{in } ^\circ\text{C}) + 32 = \frac{9}{5} (-164^\circ\text{C}) + 32 = -263.2 = \mathbf{-263^\circ\text{F}}$$

$$\text{c) } T (\text{in } ^\circ\text{C}) = T (\text{in K}) - 273.15 = 0 \text{ K} - 273.15 = -273.15 = \mathbf{-273^\circ\text{C}}$$

$$T (\text{in } ^\circ\text{F}) = \frac{9}{5} T (\text{in } ^\circ\text{C}) + 32 = \frac{9}{5} (-273.15^\circ\text{C}) + 32 = -459.67 = \mathbf{-460. ^\circ\text{F}}$$

1.45 Plan: Use $10^{-9} \text{ m} = 1 \text{ nm}$ to convert wavelength. Use $0.01 \text{ \AA} = 1 \text{ pm}$, $10^{-12} \text{ m} = 1 \text{ pm}$, and $10^{-9} \text{ m} = 1 \text{ nm}$

Solution:

$$\text{a) } 247 \text{ nm} \left(\frac{10^{-9} \text{ m}}{1 \text{ nm}} \right) = \mathbf{2.47 \times 10^{-7} \text{ m}}$$

$$\text{b) } 6760 \text{ pm} \left(\frac{0.01 \text{ \AA}}{1 \text{ pm}} \right) = \mathbf{67.6 \text{ \AA}}$$

1.58 Plan: Use a calculator to obtain an initial value. Use the rules for significant figures and rounding to get the final answer.

Solution:

a) $\frac{(2.795 \text{ m})(3.10 \text{ m})}{6.48 \text{ m}} = 1.3371 = \mathbf{1.34 \text{ m}}$ (maximum of 3 significant figures allowed since two of the original numbers in the calculation have only 3 significant figures)

b) $V = \left(\frac{4}{3}\right) \pi (17.282 \text{ m})^3 = 72.3907 = \mathbf{72.391 \text{ m}^3}$ (maximum of 5 significant figures allowed)

c) $1.110 \text{ cm} + 17.3 \text{ cm} + 108.2 \text{ cm} + 316 \text{ cm} = 442.61 = \mathbf{443 \text{ cm}}$ (no digits allowed to the right of the decimal since 316 has no digits to the right of the decimal point)

1.60 Plan: Review the procedure for changing a number to scientific notation.

Solution:

a) $\mathbf{1.310000 \times 10^5}$ (Note that all zeros are significant.)

b) $\mathbf{4.7 \times 10^{-4}}$ (No zeros are significant.)

c) $\mathbf{2.10006 \times 10^5}$

d) $\mathbf{2.1605 \times 10^3}$

1.62 Plan: Review the examples for changing a number from scientific notation to standard notation.

Solution:

a) $\mathbf{5550}$ (Do not use terminal decimal point since the zero is not significant)

b) $\mathbf{10070.}$ (Use terminal decimal point since final zero is significant)

c) $\mathbf{0.000000885}$

d) $\mathbf{0.003004}$